Improved Elastic Weight Consolidation as an Optimization Constraint for Continual Learning

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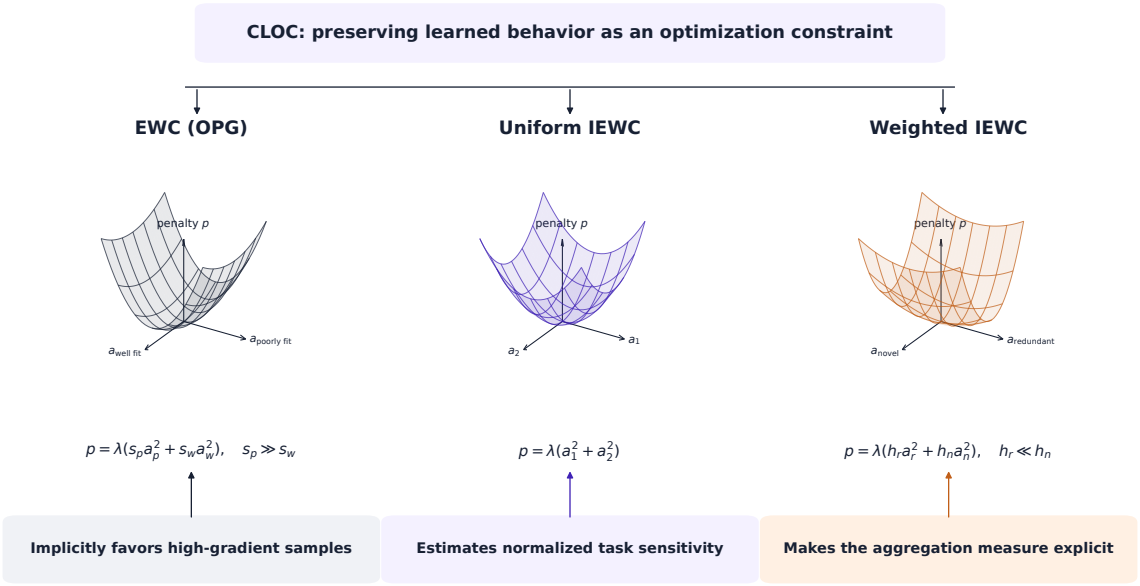


Figure 1: Each panel shows the continual-learning penalty over normalized directions for two representative samples. EWC (OPG) inherits sample weighting from output-loss-gradient scale; uniform IEWC removes this scale away from the damping regime, while residual-diversity weighting changes the aggregation measure as a separate design choice.

Abstract

Elastic Weight Consolidation (EWC) controls forgetting with a quadratic penalty, yet its empirical Fisher is usually justified through probabilistic curvature. We formulate *Continual Learning as an Optimization Constraint* (CLOC), preserving chosen functionals of previously learned behavior. First-order preservation of any differentiable scalar functional yields a gradient outer product; for per-sample negative log likelihood, it is exactly EWC's empirical Fisher, without a model-Fisher interpretation. Our central method, Improved Elastic Weight Consolidation (IEWC), instead preserves outputs modulo motions tangent to old loss level sets, separating EWC's implicit loss-gradient weighting from parameter sensitivity. IEWC is specified by $(\lambda, \kappa, \Gamma, h, \Pi)$: preservation strength, relative damping, output geometry, sample weights, and matrix approximation. A contamination experiment confirms that normalization prevents corrupt samples from dominating the full matrix. Across five domains, uniform IEWC leads EWC's selected means in classification and segmentation, while weighted IEWC leads in language and is nearly tied with EWC in forecasting and diffusion. In a validation-selected locked study, uniform IEWC improves ImageNet-R final average accuracy over matched EWC by 2.00 percentage points (95% CI 0.93–3.07) and gives no material mean difference on TRACE (−0.11; 95% CI −1.48–1.27).

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1 CONTINUAL LEARNING AS AN OPTIMIZATION CONSTRAINT

Continual learning asks a model to learn from sequentially available distributions without unrestricted access to earlier data (McCloskey and Cohen, 1989; French, 1999; Parisi et al., 2019; van de Ven and Tolias, 2019). After training on distribution A and obtaining $\theta_A^\star$, minimizing only the later loss $L_B(\theta)$ leaves previously learned behavior unconstrained. Methods such as EWC, SI, MAS, and LwF address this problem with different preservation objects: parameter importance, learning-path importance, output sensitivity, or old predictions (Kirkpatrick et al., 2017; Zenke et al., 2017; Aljundi et al., 2018; Li and Hoiem, 2016). This variety raises a prior question: *what quantity should a local continual-learning penalty preserve?*

We pose that question directly as constrained optimization. Let $q : \Theta \to \mathcal{H}$ be a quantity whose old value or acceptable set represents preservation, and let $\mathcal{C} \subset \mathcal{H}$ encode the corresponding constraint. Learning distribution B becomes

$$\min_{\theta \in \Theta} L_B(\theta) \qquad \text{subject to} \qquad q(\theta) \in \mathcal{C}.$$

We call this perspective *Continual Learning as an Optimization Constraint* (CLOC). CLOC is deliberately a formulation rather than a single algorithm: different choices of q, $\mathcal{C}$, relaxation, and approximation recover different local preservation penalties.

The paper develops one path through this design space. First, a population and low-order relaxation turns samplewise preservation objects into a stored quadratic penalty. Second, preserving any differentiable scalar functional yields a gradient outer-product matrix; choosing the old-sample loss gives OPG, and negative log likelihood gives the empirical Fisher used in common EWC implementations. Third, preserving output motion only in the normal quotient of the old loss level set yields the iEF normalization and IEWC. We make damping, sample weighting, output metric, and matrix approximation explicit design choices instead of treating them as consequences of the normalization. Finally, controlled mechanism tests and validation-selected, locked evaluations ask which parts of this construction survive finite damping and improve continual learning in the studied domains.

Concretely, the paper contributes (i) a functional-preservation construction whose loss specialization derives empirical-Fisher EWC and level-set-normalized IEWC within one framework; (ii) a matched output-metric construction and the exact finite-damping, interpolation, and loss-unit boundary; and (iii) direct spectral mechanism evidence plus a matched, validation-selected locked factorial, complemented by the retained five-domain performance record.

This framing preserves the original intuition—continual learning as preservation of previously learned behavior—while separating exact local algebra from global curvature claims and empirical questions. The next section constructs the practical relaxation and states its feasibility and locality boundaries.

2 RELATED WORK AND SCOPE

EWC stores a quadratic approximation around earlier parameters and weights parameter motion by a Fisher-derived importance estimate (Kirkpatrick et al., 2017). Subsequent work has clarified how multiple task penalties should be accumulated and placed EWC within broader Bayesian and regularization-based views of continual learning (Huszar, 2018; Schwarz et al., 2018; Nguyen et al., 2018). A practical ambiguity remains: implementations called "Fisher" may use the model-label expectation, sampled model labels, observed labels, or minibatch aggregates. These estimators are not interchangeable in theory or practice (Kunstner et al., 2019; van de Ven, 2025). Our scalar-loss derivation concerns the observed-label OPG object; sampled model-Fisher EWC is evaluated separately rather than treated as another name for the same matrix.

Other regularization methods protect different objects. SI estimates parameter importance from the optimization trajectory, MAS uses sensitivity of the learned output function, and LwF distills old predictions on later inputs (Zenke et al., 2017; Aljundi et al., 2018; Li and Hoiem, 2016). Functional regularization can instead preserve selected predictive functions or memorable points (Titsias et al., 2020; Pan et al., 2020). CLOC is complementary to this taxonomy: it begins by declaring a preservation object and constraint set, then asks which local stored penalty follows. The algebra below allows differentiable scalar functionals, while the IEWC method and experiments specialize to scalar-loss preservation and loss-level-set-normal output preservation. We do not claim that these are the only useful choices or that replay, distillation, and parameter isolation share the same resource assumptions.

iEF was introduced as a normalized observed-gradient approximation for natural-gradient optimization (Wu et al., 2024). We recover its Euclidean algebra from a different question: the metric distance of an old output displacement from the tangent to its old loss level set. This provides a preservation interpretation but does not convert iEF into exact global curvature. Our additions relative to simply substituting iEF into EWC are the CLOC derivation, the matched output-metric generalization, the finite-damping and inter-

polation boundary, and the separation of normalization from aggregation weights. The RDW ablation borrows a streaming residual-diversity intuition from GSS (Aljundi et al., 2019); it is an explicitly order-dependent heuristic, not part of the level-set result.

3 POPULATION AND LOCAL APPROXIMATIONS

For a closed convex set $\mathcal{C}$, the Moreau envelope of its indicator is a scaled squared distance (Parikh and Boyd, 2014). This motivates the soft CLOC objective

$$\min_\theta J(\theta) = \min_\theta L_B(\theta) + \frac{\lambda}{2}\operatorname{dist}(q(\theta), \mathcal{C})^2 = \min_\theta L_B(\theta) + \frac{\lambda}{2}\inf_{u\in\mathcal{C}} \|q(\theta) - u\|^2.$$

GEM and A-GEM are similar in spirit in that they formulate preservation through constraints on old examples (Lopez-Paz and Ranzato, 2017; Chaudhry et al., 2019). CLOC turns the old-task constraint into a penalty built after task-A training. In a population setting, we use sample-based analogues ℓ_B and q and take the empirical average. Define the unscaled preservation loss as $p(a, \theta) = \operatorname{dist}(q(a, \theta), \mathcal{C})^2$.

$$\min_\theta \ \frac{1}{N_B}\sum_{i=1}^{N_B} \ell_B(b_i, \theta) + \frac{\lambda}{2N_A}\sum_{i=1}^{N_A} p(a_i, \theta)$$

The convex Moreau–Yosida identity motivates this penalty form but is not a global theorem for every set used below. Neural-network loss level sets are generally nonconvex, their projections need not be unique, and squared distance need not be globally smooth. Our operative construction is local: we replace each exact set by its tangent model at the old output and derive the resulting quadratic penalty. We do not require a globally convex old-loss level set.

The samplewise population form is also stronger than preserving only expected old risk: it asks every retained sample surrogate to remain near its own constraint. Those equalities can conflict with learning the new distribution or with each other after approximation. The penalty parameter λ is therefore a soft feasibility–plasticity tradeoff, not a guarantee that all hard constraints remain exactly satisfiable.

The two-distribution presentation is only for notational simplicity; additional old distributions can be handled by adding analogous constraints. We refer to this population version as **population CLOC (P-CLOC)**. One efficiency consideration is the penalty p: Ideally, we let the constraint contribute to each gradient step with its gradient. However, recomputing the full old-distribution penalty p and its associated factors many times during optimization on distribution B is usually prohibitively expensive, since it depends on the dataset D_A associated with A. Additionally, reusing dataset D_A later on does not respect the continual learning setting. For this reason, replace the exact penalty by a local k-th order Taylor-polynomial approximation around the old solution $\theta_A^\star$. This turns the computation of p into a repeated evaluation of precomputed derivative tensors and bounds memory requirements to $O(d^k)$ for a d-dimensional parameterization. Since memory scales exponentially with k, the standard scalable surrogate is a low-order Taylor approximation. We refer to this local approximation as **low-order CLOC (L-CLOC)**, and to its population version as **PL-CLOC**. It remains to specify the preserved quantity q and constraint set $\mathcal{C}$. Notice that the gradient of p is dependent on only one sample at once, which allows us to analyze p and its gradients on a per-sample basis.

4 SCALAR FUNCTIONAL PRESERVATION

Let $s_a : \Theta \to \mathbb{R}$ be a differentiable scalar functional of the model on old sample a. Preserving that functional means

$$q(a, \theta) = s_a(\theta) - s_a(\theta_A^\star).$$

Since preservation means $q(a, \theta) = 0$, take $\mathcal{C} = \{0\}$. The corresponding unscaled proximal loss is

$$p(a, \theta) = \operatorname{dist}(q(a, \theta), \mathcal{C})^2 = q(a, \theta)^2.$$

To obtain a low-order Taylor approximation, use the perturbation around the old solution $\delta = \theta - \theta_A^\star$ in the first-order expansion of q:

$$q(a, \theta_A^\star + \delta) \approx g_a^\top \delta,$$

where

$$g_a = \nabla_\theta s_a(\theta)\big|_{\theta=\theta_A^\star}.$$

When the functional factors through the model output as $s_a(\theta) = \phi_a(f_\theta(x))$, write

$$\begin{aligned} z_{A,i} &= f_{\theta_A^\star}(x_i), \\ r_i &= \nabla_z \phi_{a_i}(z)\big|_{z=z_{A,i}}, \\ J_i &= \nabla_\theta f_{\theta_A^\star}(x_i), \end{aligned}$$

the chain rule gives

$$g_i = J_i^\top r_i.$$

Substituting this expansion into the preservation penalty gives

$$\frac{\lambda}{2N_A}\sum_{i=1}^{N_A} q(a_i, \theta)^2 \ \approx \ \frac{\lambda}{2N_A}\sum_{i=1}^{N_A} (g_i^\top \delta)^2.$$

Collecting the gradient outer products yields

$$\frac{\lambda}{2N_A}\sum_{i=1}^{N_A}(g_i^\top\delta)^2=\frac{\lambda}{2}\,\delta^\top\left(\frac{1}{N_A}\sum_{i=1}^{N_A}g_ig_i^\top\right)\delta=\frac{\lambda}{2}\,\delta^\top M_1\delta.$$

Therefore the resulting matrix is

$$M_1=\frac{1}{N_A}\sum_{i=1}^{N_A}g_ig_i^\top=\frac{1}{N_A}\sum_{i=1}^{N_A}J_i^\top r_ir_i^\top J_i.$$

This matrix encodes the local parameter directions in which the chosen scalar functional changes. The construction is exact for the squared first-order surrogate displayed above, not for the original nonlinear equality constraint. The rest of the paper takes $\phi_a=\ell_A$ unless stated otherwise. This loss specialization is the *outer-product loss-gradient* (OPG) matrix, and for negative log likelihood it becomes the empirical-Fisher algebra used by EWC, as specified next.

5 OUTER-PRODUCT LOSS-GRADIENT REGULARIZATION AND EWC

For the loss functional $s_{a_i}(\theta)=\ell_A(f_\theta(x_i))$, the matrix obtained in the previous section is

$$M_{\mathrm{OPG}}=\frac{1}{N_A}\sum_{i=1}^{N_A}g_ig_i^\top.$$

It is an outer product of loss gradients. This identity holds for any differentiable loss, without a probabilistic model or a curvature interpretation. We therefore call it an *outer-product-of-loss-gradients* (OPG) matrix.

For a negative-log-likelihood loss,

$$\ell_i(\theta)=-\log p_\theta(y_i\mid x_i),$$

we have $g_i=-\nabla_\theta\log p_\theta(y_i\mid x_i)$. The same OPG algebra then coincides with the observed-label score-gradient matrix commonly called the empirical Fisher,

$$F_{\mathrm{emp}}=\frac{1}{N_A}\sum_{i=1}^{N_A}\nabla_\theta\log p_\theta(y_i\mid x_i)\nabla_\theta\log p_\theta(y_i\mid x_i)^\top.$$

Thus OPG-EWC and empirical-Fisher EWC denote the same implementation in an explicit likelihood setting. The CLOC derivation justifies this matrix as the quadratic form of a squared first-order loss-preservation surrogate. It does not turn the matrix into the model Fisher or establish that it approximates the Hessian.

The model Fisher is instead

$$F_{\mathrm{model}}(\theta)=\mathbb{E}_{x\sim\mathcal{D}_A}\mathbb{E}_{y\sim p_\theta(\cdot|x)}\left[\nabla_\theta\log p_\theta(y\mid x)\nabla_\theta\log p_\theta(y\mid x)^\top\right].$$

It averages over labels sampled from the model, not only the observed label. Under standard regularity conditions this Fisher is connected to expected negative-log-likelihood curvature and natural-gradient geometry (Amari, 1998; Martens, 2020; Martens and Grosse, 2015). The observed-label empirical Fisher is a different object and can be a poor curvature approximation (Kunstner et al., 2019); Fisher-estimator choice can also materially affect EWC behavior and regularization scale (van de Ven, 2025).

This distinction separates two questions that are often conflated. First, why might the observed-label OPG be a useful continual-learning importance matrix? The local CLOC construction answers: it penalizes first-order changes of old sample losses. Second, is that matrix a faithful estimate of probabilistic Fisher curvature? Not in general. Our experiments therefore include a one-label Monte Carlo model-Fisher baseline, obtained by sampling from the frozen predictive distribution at each supervised output site—one class per image and one token per supervised TRACE answer position—alongside OPG-EWC.

The word "exact" below always refers to the algebra of the squared first-order local surrogate. The original hard per-sample constraint, its population counterpart, and the nonlinear loss away from $\theta_A^\star$ remain approximated.

6 LEVEL-SET PRESERVATION IN OUTPUT SPACE

Section 4 preserves a scalar functional value directly. A second construction instead preserves the model output up to motions along a level set of a differentiable output functional $\phi_a:\mathcal{Z}\to\mathbb{R}$. This output-space perspective connects to distillation and functional regularization approaches to continual learning (Li and Hoiem, 2016; Hinton et al., 2015; Titsias et al., 2020; Pan et al., 2020). Let $a\in D_A$ be an old sample, write $z_A=f_{\theta_A^\star}(x)$, and define the exact preservation set

$$\mathcal{C}_a=\{z:\phi_a(z)=\phi_a(z_A)\}.$$

Thus the preserved quantity is the model output itself:

$$q(a,\theta)=f_\theta(x).$$

The hard constraint is

$$q(a,\theta)\in\mathcal{C}_a,$$

and the corresponding unscaled proximal loss is

$$p(a,\theta)=\mathrm{dist}_G\left(q(a,\theta),\mathcal{C}_a\right)^2,$$

where $G\succ 0$ is the chosen local output metric. Positive definiteness makes the primal metric and its inverse on covectors unambiguous; Section 12 treats the

restricted pseudoinverse convention needed by a singular ordered-output example.

This exact penalty is generally not tractable. To obtain a practical form, use the standard first-order local model of the constraint set near the old output z_A, namely its tangent plane. Let

$$r = \nabla_z \phi_a(z)\big|_{z=z_A}.$$

We first assume $r \neq 0$, so the functional's level set has a regular tangent hyperplane at z_A. For a small output displacement $\Delta z = f_\theta(x) - f_{\theta_A^\star}(x)$, its first-order change is approximately $r^\top \Delta z$. Hence the local constraint set is

$$T = \{\Delta z : r^\top \Delta z = 0\}.$$

Replacing the exact level set by this tangent hyperplane gives the local proximal loss

$$p(a, \theta) \approx \mathrm{dist}_G(\Delta z, T)^2.$$

For the metric $|v|_G^2 = v^\top G v$, the squared distance to the tangent hyperplane follows from projecting Δz onto the gradient:

$$\mathrm{dist}_G(\Delta z, T)^2 = \frac{(r^\top \Delta z)^2}{r^\top G^{-1} r}.$$

To verify the matched metric pair, write the normal residual as $e = \Delta z - t$ with $t \in T$. It solves

$$\min_e e^\top G e \quad \text{subject to} \quad r^\top e = r^\top \Delta z.$$

The Lagrange equations give

$$e^\star = \frac{r^\top \Delta z}{r^\top G^{-1} r} G^{-1} r,$$

whose squared G-norm is the displayed quotient. Thus G measures primal output displacement while G^{-1} acts on the loss covector; these roles cannot be chosen independently. Linearizing the model output around $\theta_A^\star$, with

$$\delta = \theta - \theta_A^\star, \qquad J = \nabla_\theta f_{\theta_A^\star}(x),$$

we have

$$\Delta z \approx J\delta.$$

Substituting this into the local regularization term gives

$$\frac{\lambda}{2} p(a, \theta_A^\star + \delta) \approx \frac{\lambda}{2} \frac{(r^\top J \delta)^2}{r^\top G^{-1} r}.$$

Equivalently,

$$\frac{\lambda}{2} p(a, \theta_A^\star + \delta) \approx \frac{\lambda}{2} \delta^\top \left(J^\top \frac{r r^\top}{r^\top G^{-1} r} J \right) \delta.$$

Taking the empirical average over old-distribution samples $a_1, \ldots, a_N$ yields an importance matrix of the form

$$M_{2,A} = \frac{1}{N} \sum_{i=1}^{N} J_i^\top \frac{r_i r_i^\top}{r_i^\top G_i^{-1} r_i} J_i.$$

This is a local construction: it penalizes motion normal to the tangent model, not distance to a globally convex neural-network level set. IEWC is the specialization $\phi_a = \ell_A$ studied below; other functionals define CLOC extensions but are not evaluated here. When $r_i = 0$, the tangent normal is undefined and the undamped quotient is $0/0$; our finite-damping implementation assigns that sample zero contribution. Section 11 makes the resulting near-interpolation boundary explicit.

7 LEVEL-SET NORMALIZATION AND THE IMPROVED EMPIRICAL FISHER

For the loss-functional specialization $\phi_a = \ell_A$ and Euclidean output metric $G_i = I$, the matrix from the previous section is the improved empirical Fisher (iEF) introduced by Wu et al. (Wu et al., 2024):

$$M_{2,I} = F_{\mathrm{IEF}}.$$

Hence iEF is the quadratic form of a first-order local constraint that protects old behavior geometrically: instead of preserving a raw scalar-loss coordinate, it penalizes only output motion normal to the old loss level set. The original iEF paper derives the same normalization for approximate natural-gradient descent. In that setting, the observed-label OPG reweights sample contributions by squared output-gradient magnitude; iEF removes this scale factor and retains the normalized output-loss direction. Here the normalization arises from metric projection onto the normal quotient of the old-loss tangent space.

This is an identity for the undamped local summand when $r_i \neq 0$. Practical damping changes its behavior near $r_i = 0$; Section 11 states that boundary explicitly.

8 RELATION TO GAUSS-NEWTON

Consider the scalar least-squares setting in Euclidean geometry. Suppose the model output is scalar, such that

$$\ell_n(z) = \frac{1}{2}(z - y_n)^2.$$

Then

$$r_n = z_n^A - y_n$$

is scalar, and for $r_n \neq 0$ we have

$$\frac{r_n^2}{r_n^2} = 1.$$

Therefore the *undamped* Euclidean IEF contribution reduces to

$$J_n^\top J_n,$$

and averaging gives

$$M_{\mathrm{IEF}} = \frac{1}{N}\sum_{n=1}^{N} J_n^\top J_n.$$

This is exactly the Gauss–Newton or output-Jacobian form for squared-error regression (Schraudolph, 2002; Martens, 2010, 2020), the standard practical preconditioning matrix for second-order optimization. This matches our geometric intuition: loss preservation is governed by how parameters move the output, which is encoded by the Jacobian J, not by the current output-gradient magnitude. This equivalence was made explicit in (Wu et al., 2024). With positive damping, the contribution is instead multiplied by $r_n^2/(r_n^2+\tau)$ and tends to zero at exact interpolation; the exact Gauss–Newton identity therefore requires $\tau = 0$ and $r_n \neq 0$.

9 NORMALIZATION AND SAMPLE WEIGHTING

To compare both matrices in a common notation, let $G_n \succ 0$ denote the chosen output metric for sample n. Although M_{OPG} itself does not depend on G_n, its summands can be rewritten relative to that metric. In a negative-log-likelihood setting, $M_{\mathrm{OPG}} = F_{\mathrm{emp}}$.

Define

$$\hat{r}_n^G = \frac{r_n}{\sqrt{r_n^\top G_n^{-1} r_n}}, \qquad a_n^G = J_n^\top \hat{r}_n^G, \qquad w_n^G = r_n^\top G_n^{-1} r_n.$$

Then, wherever $r_n \neq 0$,

$$M_{\mathrm{IEF},G} = \frac{1}{N}\sum_{n=1}^{N} a_n^G (a_n^G)^\top, \qquad M_{\mathrm{OPG}} = \frac{1}{N}\sum_{n=1}^{N} w_n^G\, a_n^G (a_n^G)^\top.$$

Both matrices empirically aggregate the same local old-task sensitivity directions $a_n^G(a_n^G)^\top$, but they differ in how strongly each sample contributes. Undamped IEF keeps uniform empirical weighting of these normalized directions, while OPG multiplies each contribution by the metric-dependent squared magnitude

$$w_n^G = r_n^\top G_n^{-1} r_n.$$

Thus normalization changes sample aggregation: OPG shifts old-task weighting toward samples with large output-loss gradients and away from confidently fitted ones, whereas undamped IEF removes that scale. With positive damping, IEF also multiplies each direction by $s_n/(s_n+\tau)$, as Section 11 makes explicit.

This decomposition identifies a mechanism to test; it does not show that either weighting is preferable for every task. The next section treats probabilistic classification, where the scale difference has a concrete well-fitted limit.

10 GRADIENT-SCALE EFFECTS IN CLASSIFICATION

The output-gradient weighting in observed-label OPG has an important classification symptom. For cross-entropy with softmax probabilities p_n and one-hot label y_n, the output-loss gradient with respect to logits is

$$r_n = p_n - y_n.$$

For a confidently correct prediction with class c and small ε,

$$p_{n,c} = 1-\varepsilon, \qquad \|r_n\|_2^2 \in \Theta(\varepsilon^2).$$

Thus the OPG contribution becomes small as the model becomes more confident and correct. This is coherent with its derivation: the scalar first-order loss change is also small. For continual-learning preservation, however, small current loss need not mean that losing the sample later is harmless. EWC-DR identifies this classification symptom and proposes a logit-reversal repair (Liu and Chang, 2026).

In the present framework, that observation is one instance of the weighting effect identified in Section 9: observed-label OPG importance depends on output-gradient scale. Undamped IEF removes this particular factor. Positive damping retains the removal only above its transition scale, and empirical benefit remains a question for the controlled comparisons below.

11 BEHAVIOR NEAR WELL-FITTED SOLUTIONS

The output-loss-gradient scale in OPG is especially consequential because importance is estimated *after* training on A, when many samples may already be well fit. The original EWC Laplace argument likewise expands at an old-task optimum, although its model-Fisher/Hessian motivation is distinct from the observed-label OPG object analyzed here (Kirkpatrick et al., 2017; Kunstner et al., 2019). Modern expressive models can fit training data up to interpolation (Belkin

et al., 2019), so the near-zero regime must be treated explicitly rather than hidden by notation.

To make the comparison precise, consider the per-sample summands

$$S_n^{\mathrm{OPG}} = J_n^\top r_n r_n^\top J_n, \qquad S_n^{\mathrm{IEF},G} = J_n^\top \frac{r_n r_n^\top}{r_n^\top G_n^{-1} r_n} J_n,$$

where $G_n \succ 0$ is the chosen output metric. We assume that G_n is uniformly positive definite and bounded, and that J_n is bounded under the operator norm. More precisely, assume there exists a constant $C_J < \infty$ such that

$$\|J_n^\top v\|_2 \le C_J \|v\|_2$$

for the relevant directions, and constants $0 < m_G \le M_G < \infty$ such that G_n is uniformly bounded between $m_G I$ and $M_G I$.

Using the same G_n-normalized direction as above,

$$\hat{r}_n^G = \frac{r_n}{\sqrt{r_n^\top G_n^{-1} r_n}}, \qquad a_n^G = J_n^\top \hat{r}_n^G, \qquad w_n^G = r_n^\top G_n^{-1} r_n, \qquad b_n^G = \|a_n^G\|_2,$$

which gives

$$S_n^{\mathrm{IEF},G} = a_n^G (a_n^G)^\top, \qquad S_n^{\mathrm{OPG}} = w_n^G \, a_n^G (a_n^G)^\top.$$

For the spectral norm,

$$\|S_n^{\mathrm{OPG}}\|_2 = w_n^G b_n^G, \qquad \|S_n^{\mathrm{IEF},G}\|_2 = b_n^G.$$

Since

$$\|\hat{r}_n^G\|_2^2 = \frac{\|r_n\|_2^2}{r_n^\top G_n^{-1} r_n},$$

the metric bounds and bounded-Jacobian assumption give $b_n^G \in O(1)$, so

$$\|S_n^{\mathrm{OPG}}\|_2 \in O(w_n^G), \qquad \|S_n^{\mathrm{IEF},G}\|_2 \in O(1).$$

The sharper lower-bound statement requires a nondegeneracy condition in the normalized output-gradient direction. If there exists $c_J > 0$ such that

$$\|J_n^\top \hat{r}_n^G\|_2 \ge c_J$$

for the relevant samples, then

$$\|S_n^{\mathrm{OPG}}\|_2 \in \Theta(w_n^G), \qquad \|S_n^{\mathrm{IEF},G}\|_2 \in \Theta(1).$$

We now interpret this for practical situations. Near the end of training on distribution A, we expect many old samples to have small gradients. The norm analysis above then shows that OPG contains the additional multiplicative factor $w_n^G = r_n^\top G_n^{-1} r_n$, whereas undamped IEF removes that factor. Remaining low-magnitude summands of the IEF come from the model sensitivity term b_n^G, not from the scale of the output-loss gradient itself. In the classification setting with $G_n = I$, a confidently correct prediction with $p_{n,c} = 1 - \varepsilon$ yields

$$\|r_n\|_2^2 = \|p_n - y_n\|_2^2 \in \Theta(\varepsilon^2).$$

Therefore

$$\|S_n^{\mathrm{OPG}}\|_2 = w_n^I b_n^I \in \Theta(\varepsilon^2)\, b_n^I, \qquad \|S_n^{\mathrm{IEF},I}\|_2 = b_n^I.$$

Under the same nondegeneracy condition on $J_n^\top \hat{r}_n^I$, this gives $\|S_n^{\mathrm{OPG}}\|_2 \in \Theta(\varepsilon^2)$ and $\|S_n^{\mathrm{IEF},I}\|_2 \in \Theta(1)$.

The singular case $r_n = 0$ remains and must be handled explicitly. As done in (Wu et al., 2024), one practical solution is to add a damping factor τ to the quotient, yielding

$$M_{\mathrm{IEF},G} = \frac{1}{N} \sum_{n=1}^{N} \frac{g_n g_n^\top}{r_n^\top G_n^{-1} r_n + \tau}, \qquad g_n = J_n^\top r_n.$$

For $s_n = r_n^\top G_n^{-1} r_n > 0$, each damped summand admits the exact factorization

$$S_{n,\tau}^{\mathrm{IEF},G} = \frac{g_n g_n^\top}{s_n + \tau} = \underbrace{\frac{s_n}{s_n + \tau}}_{\rho_\tau(s_n)} S_{n,0}^{\mathrm{IEF},G}.$$

Thus finite damping introduces a transition rather than a uniform scale removal. When $s_n \gg \tau$, $\rho_\tau(s_n) \approx 1$ and the undamped direction-normalization analysis applies. When $s_n \ll \tau$, $\rho_\tau(s_n) \approx s_n/\tau$, so small output-loss gradients are again suppressed. Since $g_n = J_n^\top r_n$, fixed positive damping gives $S_{n,\tau}^{\mathrm{IEF},G} \to 0$ as $r_n \to 0$ under a bounded Jacobian. At exact interpolation, $r_n = g_n = 0$; we define the sample contribution as zero. The undamped expression is instead $0/0$ and has no canonical value without specifying a directional limit.

Fixed numerical τ also has the units of s_n. Under a global rescaling of one task loss by c, the numerator and s_n scale by c^2, but a fixed τ does not. We therefore use task-relative damping in the confirmatory experiments:

$$\tau_A = \kappa \ \mathrm{median}\{s_i : s_i > 0\}, \qquad \kappa = 10^{-2}.$$

Both s_n and τ_A then scale by c^2, giving

$$\frac{c^2 g_n g_n^\top}{c^2 s_n + c^2 \tau_A} = \frac{g_n g_n^\top}{s_n + \tau_A}.$$

Task-relative damping therefore restores invariance to global changes of loss units while retaining explicit attenuation near zero gradient. It does not imply invariance to arbitrary sample-dependent monotone transformations under a shared positive damping value. If all recorded s_i are zero, the task importance is defined as the zero matrix.

12 OUTPUT-SPACE GEOMETRY

The scalar CLOC construction and its OPG matrix require no output-space metric. The level-set construction does: for a symmetric positive-definite primal metric G_n, output displacements have inner product $\langle u, v\rangle_{G_n} = u^\top G_n v$, the metric-normal direction is $G_n^{-1} r_n$, and the normal-distance denominator is $r_n^\top G_n^{-1} r_n$. Thus a geometry choice must supply a matched pair: G_n acts on primal output displacement and G_n^{-1} on the loss covector. An arbitrary denominator does not constitute a new geometry.

The canonical choice $G_n = I$ recovers iEF (Wu et al., 2024). More generally, any smooth output discrepancy whose second-order expansion supplies a valid local quadratic metric can enter IEWC through its matched primal/dual pair G_n, G_n^{-1}. For singular ambient metrics, this requires a stated tangent space or quotient, Moore–Penrose pseudoinverse, and range conditions; diagonalizing the parameter pullback can further discard cross-parameter geometry.

For ordered probability outputs, Appendix G constructs a CDF/Cramér metric and its matched pseudoinverse dual on the simplex tangent. This preserves the original motivation for space-dependent output geometry but is not sliced Wasserstein. The earlier diffusion experiment therefore remains exploratory evidence for output-representation flexibility rather than validation of a Wasserstein geometry.

13 NOTATION AND APPROXIMATION BOUNDARIES

Table 1 collects the objects that are otherwise easy to conflate.

Our derivations concern full positive-semidefinite matrices, whereas the training experiments store only their diagonals. Taking $\mathrm{diag}(M)$ preserves each coordinate-axis quadratic penalty, nonnegativity of the entries, and the trace. It generally does not preserve off-axis quadratic forms, rank, eigenvectors, parameterization behavior, or the full normal-space geometry. Consequently, the full-matrix contamination analysis in Appendix B tests the proposed normalization mechanism, while benchmark training separately tests whether its diagonal projection is useful.

Appendix A gives the exact scope of loss-rescaling invariance and Appendix B analyzes and directly tests the contamination mechanism.

Table 1: Notation and distinctions used throughout the paper.

Symbol or name	Meaning
$g_i = J_i^\top r_i$	Parameter gradient of the selected scalar functional; IEWC uses the old-sample loss.
M_{OPG}	Mean $g_i g_i^\top$ from scalar-loss preservation; only the likelihood-score case is the observed-label empirical-Fisher algebra.
F_{model}	Fisher information averaging score outer products over labels sampled from the frozen predictive distribution.
$s_i = r_i^\top G_i^{-1} r_i$	Squared dual norm of the output-loss gradient under the matched output metric.
M_{IEF}	Level-set-normalized mean $g_i g_i^\top/(s_i + \tau_A)$; the undamped Euclidean case is iEF.
$\rho_\tau(s)$	Damping attenuation $s/(s+\tau)$ relative to the undamped summand.
κ	Dimensionless relative-damping coefficient in $\tau_A = \kappa\,\mathrm{median}_{s_i>0}\, s_i$.
$\bar{h}_i$	Optional mean-one aggregation weight; uniform weighting and RDW are evaluated separately.
(Γ, h, Π)	Output-metric rule, aggregation-weight rule, and matrix approximation operator.

14 IMPROVED ELASTIC WEIGHT CONSOLIDATION

The preceding derivation leaves five implementation choices: global penalty strength, damping, aggregation weights, output geometry, and matrix approximation. For old-task samples $a_1, \ldots, a_N$, IEWC uses

$$M_{\mathrm{IEWC}} = \frac{1}{N}\sum_{n=1}^{N} \bar{h}_n \frac{g_n g_n^\top}{r_n^\top G_n^{-1} r_n + \tau_A}, \qquad g_n = J_n^\top r_n,$$

where $\bar{h}_n \geq 0$ has empirical mean one. The continual-learning penalty is

$$\frac{\lambda}{2}(\theta - \theta_A^\star)^\top M_{\mathrm{IEWC}}(\theta - \theta_A^\star).$$

The canonical geometry is $G_n = I$. We use task-relative damping

$$\tau_A = \kappa\ \mathrm{median}\{r_i^\top G_i^{-1} r_i : r_i \neq 0\},$$

with $\kappa = 10^{-2}$ in the confirmatory comparison. This makes κ dimensionless and preserves global loss-unit invariance. If all denominators are zero, the stored importance is zero.

Uniform IEWC sets $\bar{h}_n = 1$. To study weighting without confounding it with normalization, we define a separate *residual-diversity weighting* (RDW) variant. Let

$$u_n = \frac{g_n}{\|g_n\|_2}$$

for nonzero parameter gradients. In a fixed sample order, RDW maintains an orthonormal basis Q of at most K earlier residual directions and assigns

$$h_n = \|(I - Q^\top Q)u_n\|_2^2.$$

When the residual is nonzero and the cap is not reached, its normalized direction is appended to Q; the final h_n are mean-normalized. This streaming rule is motivated by residual-gradient diversity ideas in GSS (Aljundi et al., 2019), but it is an order-dependent aggregation heuristic, not part of the level-set derivation. Crucially, both OPG-RDW and IEF-RDW receive the same unit parameter-gradient directions. Scalar IEF normalization cannot change u_n, so the 2×2 comparison

$$\{\text{OPG}, \text{IEF}\} \times \{\text{uniform}, \text{RDW}\}$$

isolates matrix normalization from sample weighting.

For diagonal training penalties, we take the diagonal after aggregation and rescale every nonzero diagonal to have mean one over trainable parameters. This removes only a global factor absorbable into λ and lets all regularized methods use the same within-domain λ grid. It does not restore the cross-parameter information discarded by diagonalization.

An IEWC instantiation is specified by the five-tuple

$$(\lambda, \kappa, \Gamma, h, \Pi),$$

where λ sets preservation strength, κ sets relative damping, Γ maps an old output to its local metric G_n, h supplies aggregation weights, and Π is the matrix approximation operator (for example, full, diagonal, or low rank). These are genuine design choices rather than values fixed by the derivation. Scalar loss preservation gives OPG-EWC; loss-level-set normal preservation gives uniform IEWC. We call this CLOC-derived mechanism *Improved Elastic Weight Consolidation* while treating its individual axes separately in theory and experiment.

15 EMPIRICAL EVALUATION

We use two complementary evidence tiers. The retained multi-domain study establishes IEWC's attained performance across classification, NLP, forecasting, segmentation, and diffusion; its selected endpoints are descriptive because they participated in method development. A later study freezes selection before locked-test access and estimates paired held-out effects. It compares IEF with OPG under matched uniform weighting and diagonal scale, then crosses OPG/IEF with uniform/RDW aggregation. Appendix C specifies data roles, fixed validation grids, repetitions, baselines, and uncertainty calculations.

Unless stated otherwise, all stored matrices are diagonal. Paper-facing EWC, IEWC, and IEWC+RDW denote observed-label OPG, uniform IEF, and residual-diversity-weighted IEF, respectively. RDW is order dependent, so its gains cannot be attributed to normalization alone; Appendix C maps the archived labels and complete protocols.

15.1 Established multi-domain results

Table 2 shows that uniform IEWC attained higher selected means than OPG-EWC on ImageNet-R and VOC 15–5, while the weighted IEWC configuration attained the highest TRACE score and the lowest forecasting and DDPM MSE. Together, these results demonstrate broad applicability and strong selected performance. The matched factorial and locked evaluation then isolates which gains generalize and which arise from normalization versus weighting.

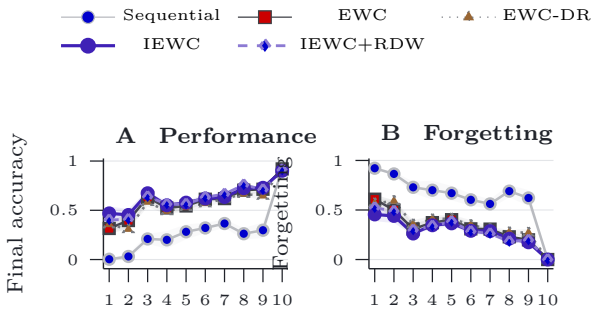


Figure 2: ImageNet-R retention in the original multi-domain study. The left panel shows final accuracy per distribution; the right shows forgetting.

At the distribution level, Figure 2 shows that IEWC's retained accuracy is not concentrated in a single late distribution. The locked paired analysis tests whether this advantage generalizes across newly frozen class orders.

Table 2: Original multi-domain continual-learning results, reported as selected-run mean (across-seed standard deviation). These development-selected endpoints establish attained performance across five task types rather than selection-independent pairwise effects. Higher is better for accuracy, score, and mIoU; lower is better for MSE and forgetting.

Task	Metric	Seq.	EWC (OPG)	EWC-DR	IEWC	IEWC+RDW
Classification	Final avg. acc. ↑	0.2896(.0405)	0.5981(.0337)	0.5687(.0263)	**0.6332(.0232)**	0.6250(.0263)
	Avg. forgetting ↓	0.7066(.0466)	0.3595(.0378)	0.3933(.0286)	**0.3142(.0250)**	0.3285(.0278)
NLP	Final avg. score ↑	0.5760(.0027)	0.5972(.0020)	0.5972(.0059)	0.5938(.0034)	**0.6272(.0080)**
	Avg. forgetting ↓	0.1139(.0111)	0.0762(.0034)	0.0781(.0109)	0.0820(.0122)	**0.0046(.0079)**
Forecasting	Final avg. MSE ↓	0.2733(.0065)	0.2388(.0021)	–	0.2418(.0038)	**0.2383(.0007)**
	MSE forgetting ↓	0.0535(.0081)	**0.0140(.0020)**	–	0.0191(.0047)	0.0147(.0021)
Segmentation	Final all mIoU ↑	0.0917(.0061)	0.0946(.0040)	0.3080(.0307)	**0.4678(.0568)**	0.4274(.0574)
	Base forgetting ↓	0.7135(.0114)	0.7131(.0166)	0.4471(.0447)	**0.2418(.0965)**	0.2925(.0839)
Diffusion	Final denoise MSE ↓	0.0676(.0029)	0.0296(.0013)	–	0.0298(.0014)	**0.0293(.0013)**
	MSE forgetting ↓	0.0503(.0002)	**-0.0006(.0020)**	–	0.0001(.0024)	-0.0003(.0020)

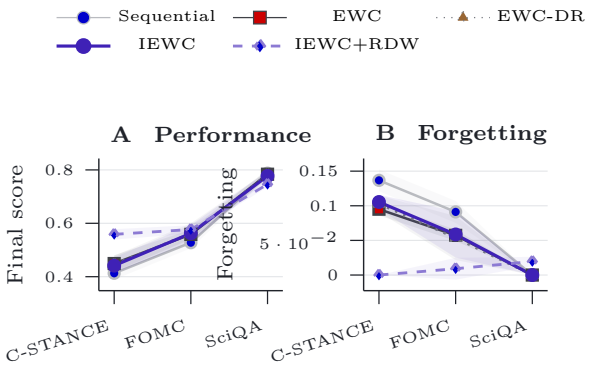


Figure 3: TRACE retention for C-STANCE, FOMC, and ScienceQA in the original study. FOMC's distributed evaluation and test files are duplicates, so the new locked comparison uses C-STANCE, ScienceQA, and NumGLUE-cm.

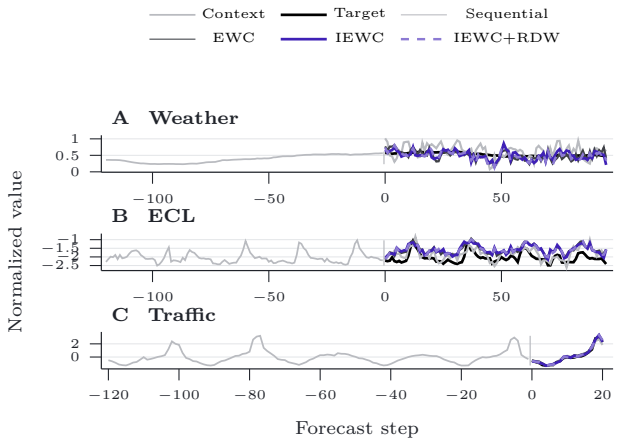


Figure 4: Forecasting rollout comparison. Each panel shows one held-out rollout from the corresponding forecasting distribution. The gray curve is the observed context, the black curve is the target horizon, and colored curves are method predictions after the full continual sequence.

The weighted IEWC run likewise has less forgetting on C-STANCE and FOMC in Figure 3. The revised comparison retains the weighting mechanism but crosses it with both OPG and IEF, while replacing FOMC with NumGLUE-cm to obtain disjoint validation and test roles.

Figure 4 complements the aggregate forecasting scores with individual held-out rollouts. Weighted IEWC retains more target-trajectory structure at the first task and remains close to OPG-EWC later; these examples illustrate the observed behavior rather than replace the aggregate uncertainty estimates.

The DDPM study also tested an alternative output representation originally labeled "sliced Wasserstein." The selected run reduced old-output drift to 9.4409 ± 7.2517, compared with 14.1026 ± 12.2220 for Euclidean uniform IEWC, while attaining denoising MSE 0.0301 ± 0.0012 versus 0.0298 ± 0.0014. Because that surrogate did not establish a matched primal/dual metric pair, the result supports output-

representation flexibility but not a claim about Wasserstein geometry. Appendix G supplies a correct ordered-CDF/Cramér construction.

The FROMP-derived weighting ablation (Pan et al., 2020) reached 0.6365 ± 0.0165 final average accuracy and 0.3068 ± 0.0170 forgetting on ImageNet-R, but 0.3351 ± 0.0742 final all-mIoU and 0.4064 ± 0.1122 base-class forgetting on VOC 15–5. Its domain-dependent behavior further motivates treating aggregation as an explicit design axis rather than folding it into the normalization claim.

Appendices B–D report the mechanism, protocol/sensitivity, and label-noise analyses; Appendix F gives the archive and reproduction path.

16 LIMITATIONS

CLOC is a formulation of preservation, not a claim that all old-task behavior can be retained. Per-sample equality is stronger than expected-risk preservation and can conflict with fitting later tasks. The penalty coefficient therefore controls a stability–plasticity relaxation rather than guaranteeing feasibility of the original hard constraint.

IEWC, like EWC, compresses an old task into one importance matrix evaluated at $\theta_A^\star$ and keeps it fixed during later training. Before aggregation, however, its normalized parameter-sensitivity direction $a^G(x;\theta_A^\star) = J_{\theta_A^\star}(x)^\top \hat{r}^G(x)$ is input-dependent. Circuit analyses make the same distinction between a model's fixed connections and the input-dependent feature contributions that realize a particular computation (Dunefsky et al., 2024). A single average—especially after diagonalization—can therefore hide sensitivity confined to subsets of inputs or distributed across overlapping pathways, and cannot track changes in the relevant Jacobians as the parameters move away from $\theta_A^\star$. Sample weighting changes which empirical average is stored, but cannot recover this conditional structure after compression. Conditional mixtures or dynamically refreshed importance are natural extensions, at the cost of richer stored summaries or access to old inputs.

The output metric must be specified as a matched primal/dual pair. Ambient singular metrics require restriction to an identifiable subspace and a stated pseudoinverse convention. Our ordered-CDF construction satisfies that requirement on the probability-simplex tangent, but the inherited diffusion result is exploratory and does not validate a general Wasserstein geometry.

Practical IEWC also introduces approximations beyond the derivation. A diagonal retains coordinatewise importance but discards correlations, eigenspaces, and reparameterization behavior. Importance estimates use finitely many old samples. RDW is an order-dependent weighting heuristic with additional cost and is not implied by level-set normalization. Relative damping guarantees invariance to a global change of loss units, not to arbitrary sample-dependent transformations at finite damping.

Finally, empirical comparisons cover selected datasets, model families, adaptation regimes, and task orders. Validation selection and locked paired evaluation reduce selection bias but cannot establish a domain-independent advantage or deployment robustness. Offline/joint training is only a reference because it violates the continual data-access constraint, and untested replay, distillation, and parameter-isolation methods may be preferable when their storage or architectural assumptions are allowed.

17 CONCLUSION

We formulated continual learning as preservation of selected functionals of previously learned behavior under subsequent optimization (CLOC) and derived two quadratic relaxations. Any differentiable scalar functional yields a gradient outer-product penalty; choosing the old-sample loss gives OPG, and negative log likelihood gives the empirical Fisher used in many EWC implementations without requiring a model-Fisher interpretation. Our central method, IEWC, instead preserves outputs modulo the tangent to the old loss level set and yields the iEF normalization. This retains the central geometric intuition while locating its exact boundary: undamped normalization removes the samplewise output-loss-gradient scale, whereas positive damping multiplies each summand by $s/(s+\tau)$ and assigns zero contribution at exact interpolation.

Task-relative damping restores invariance to global changes of loss units, and the controlled contamination study directly shows that normalization can prevent high-loss corrupted samples from dominating a full importance matrix in the tested construction. Whether this altered local geometry improves a diagonal continual-learning penalty remains an empirical, domain-conditional question. Our evaluation therefore separates normalization from residual-diversity weighting, matches global scale, includes external regularization and offline references, selects hyperparameters only on validation data, and reserves paired task orders and test splits for locked inference. IEWC is therefore a preservation-first CLOC family: its explicit damping, weighting, output metric, and matrix approximation explain normalized importance geometrically and guide new continual-learning penalties.

AI Use Statement

The original IEWC framework and initial manuscript predate the generative-AI assistance disclosed here. For the present revision, OpenAI Codex language models assisted with review-response planning; literature search and summarization; checking, extending, and editing mathematical derivations; designing the frozen confirmatory protocol; creating and editing experiment, analysis, testing, and archival code; experiment orchestration and failure diagnosis; generating draft tables and figures; interpreting preserved result records; and drafting and editing manuscript text. Generative AI was not used to create benchmark labels or examples, alter evaluation data, select outcomes after locked-test access, conduct human-subject analysis, translate material, or transcribe recordings.

The author retained responsibility for the research questions, protocol and claim decisions, and final manuscript. AI-assisted code was checked through unit and integration tests, deterministic duplicate runs, clean-checkout reproduction, and artifact checksums. Numerical claims were regenerated from preserved records and checked against machine-readable summaries; mathematical claims were checked against the stated assumptions and displayed derivations. The author takes responsibility for all content of the submission, including AI-assisted text, code, analyses, and artifacts.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
   (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes. Sections 1–13 derive the construction and Appendix F specifies the executable methods.]
   (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes. Sections 10–13 analyze the construction; Appendix F states storage and importance-estimation costs and Appendix C reports measured time and peak memory.]
   (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes. The supplementary archive contains both the frozen run source and the reproduction source, together with a pinned lockfile.]

2. For any theoretical claim, check if you include:
   (a) Statements of the full set of assumptions of all theoretical results. [Yes. The assumptions appear with the corresponding derivations in Sections 2, 5, 10, and 11 and Appendix A.]
   (b) Complete proofs of all theoretical results. [Yes. The results used by the method are proved algebraically in the main text and Appendix A.]
   (c) Clear explanations of any assumptions. [Yes. Sections 2, 10–12, and the limitations section explain their operational consequences.]

3. For all figures and tables that present empirical results, check if you include:
   (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes. The supplementary archive contains source, analysis-complete records, manifests, checksums, and end-to-end instructions; datasets and checkpoints are obtained from their cited sources.]
   (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes. Appendix C records the complete frozen protocol, selection rule, roles, grids, and training configurations.]
   (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes. Section 14 and Appendix C define the metrics, repetitions, paired intervals, randomization tests, and dependence checks.]
   (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes. Appendix F reports the single-workstation CPU/GPU environment and the measured resource fields.]

4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
   (a) Citations of the creator If your work uses existing assets. [Yes.]
   (b) The license information of the assets, if applicable. [Yes. Appendix F states the applicable upstream terms and redistribution boundary.]
   (c) New assets either in the supplemental material or as a URL, if applicable. [Yes. The anonymized supplementary archive contains the new protocol, result records, analyses, and source snapshots.]
   (d) Information about consent from data providers/curators. [Not Applicable. The work uses existing public research benchmarks and does not collect new data from people.]
   (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable to the new artifacts. No benchmark examples are redistributed, and restricted prompt and target fields are removed from archived run records.]

5. If you used crowdsourcing or conducted research with human subjects, check if you include:

(a) The full text of instructions given to participants and screenshots. [Not Applicable.]

(b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable.]

(c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable.]

Improved Elastic Weight Consolidation as an Optimization Constraint for Continual Learning: Supplementary Materials

A INVARIANCE TO LOSS REPARAMETERIZATION

This appendix treats the undamped case $\tau = 0$.

The output-level feasible set

$$\mathcal{C}_n = \{z : \ell_n(z) = \ell_n(z_n^A)\}$$

depends on the level sets of ℓ_n, not on the numerical labels assigned to those level sets. If we apply a local monotone scalar transformation

$$\tilde{\ell} = \phi(\ell), \qquad \phi'(\ell) > 0,$$

then the output gradient rescales as

$$\tilde{r}_n = \phi'(\ell_n(z_n^A))\, r_n.$$

For OPG, the sample contribution changes by

$$J_n^\top \tilde{r}_n \tilde{r}_n^\top J_n = \phi'(\ell_n(z_n^A))^2 J_n^\top r_n r_n^\top J_n.$$

For IEF, the same scalar factor appears in numerator and denominator:

$$\frac{\tilde{r}_n \tilde{r}_n^\top}{\tilde{r}_n^\top G_n^{-1} \tilde{r}_n} = \frac{r_n r_n^\top}{r_n^\top G_n^{-1} r_n}.$$

Thus undamped IEF is invariant to local scalar relabelings of the constraint function wherever $r_n \neq 0$. This includes loss rescaling, unit changes in regression losses, reduction conventions, and smooth monotone transformations that preserve the local level-set geometry. OPG's global scale can be absorbed into λ only when the transformation rescales all samples equally; sample-dependent factors change its relative weighting.

Positive shared damping narrows the invariance. A fixed numerical τ does not transform with the loss and therefore changes even under a global unit rescaling. Task-relative damping $\tau_A = \kappa \,\mathrm{median}_{i:s_i>0}\, s_i$ restores invariance to a global task-loss multiplier because s_i and τ_A scale together. It does not generally preserve invariance to sample-dependent monotone transformations. At $r_n = 0$, the implementation assigns a zero contribution rather than claiming an undamped 0/0 identity.

B CONTAMINATION CONTRIBUTIONS AND SPECTRAL STABILITY

The OPG–IEF normalization difference can change which samples dominate a full importance matrix. In Euclidean output geometry, write $s_i = \|r_i\|_2^2$ and $a_i = g_i/\sqrt{s_i}$ for $s_i > 0$. The undamped per-sample matrices are

$$M_{\mathrm{OPG},i} = s_i a_i a_i^\top, \qquad M_{\mathrm{IEF},i} = a_i a_i^\top.$$

Thus OPG couples each sensitivity direction to the squared output-loss-gradient scale, whereas undamped IEF keeps the direction-only outer product. Relative damping multiplies IEF by $s_i/(s_i + \tau_A)$ and therefore retains the near-zero qualification from Section 11.

For intuition, suppose a fraction ε of samples are contaminated, their normalized sensitivity is concentrated near a direction u outside the leading clean subspace, and their scale is $s_i \approx W$. If clean scales are sufficiently concentrated or independent of direction, then

$$\mathbb{E}[M_{\mathrm{OPG}}] \approx (1-\varepsilon)\bar{s}_{\mathrm{clean}}\Sigma_{\mathrm{clean}} + \varepsilon W u u^\top,$$

whereas undamped IEF has no factor W on the contaminating direction. In particular, a sufficiently large εW can move u into the leading OPG eigenspace. This model is illustrative rather than a distribution-free robustness theorem.

We test the mechanism directly in a fitted two-class MLP. For each seed, 512 inputs are drawn from $\mathcal{N}(0, I_2)$ and assigned the clean label $\mathbf{1}\{x_1 + 0.75x_2 < 0\}$; a uniformly selected 0, 5, 10, or 20% of labels is flipped. A 2–32–2 tanh MLP is trained for 250 epochs with Adam, learning rate 0.01, and batch size 64. For each of 20 independent seeds we then retain every per-sample parameter-gradient row and clean/corrupt identity. We decompose the full OPG and task-relative-damped IEF matrices ($\kappa = 0.01$) into clean and corrupt contributions, then compare each total matrix with its clean-contribution reference. Table 3 reports means over seeds.

Table 3: Direct full-matrix contamination diagnostic. Alignment is the absolute inner product of the leading eigenvectors; top-5 overlap is the mean squared singular value between the corresponding eigenspaces. Each reference is the clean contribution from the same fitted model.

Flip rate	Corrupt/clean s_i ratio	Corrupt trace share		Leading-vector alignment		Top-5 overlap	
		OPG	IEF	OPG	IEF	OPG	IEF
0.05	81.26	0.783	0.054	0.426	1.000	0.902	0.996
0.10	36.67	0.789	0.104	0.720	0.999	0.923	0.997
0.20	11.23	0.736	0.202	0.875	0.998	0.930	0.990

Table 4: Paired mean differences with 95% Student-t intervals over 20 seeds. Every interval excludes zero.

IEF minus OPG contrast	5% flips	10% flips	20% flips
Corrupt trace share	-0.729 $[-0.751, -0.706]$	-0.685 $[-0.701, -0.670]$	-0.534 $[-0.549, -0.519]$
Leading-vector alignment	$+0.574$ $[+0.444, +0.704]$	$+0.279$ $[+0.152, +0.406]$	$+0.123$ $[+0.034, +0.213]$
Top-5 overlap	$+0.094$ $[+0.067, +0.122]$	$+0.074$ $[+0.047, +0.101]$	$+0.060$ $[+0.029, +0.091]$

These observations support the proposed contamination mechanism in this controlled construction. They do not establish a universal robustness guarantee or prove that a diagonal continual-learning penalty improves on a real benchmark; those are separate empirical questions.

C EXPERIMENTAL PROTOCOL, PROVENANCE, AND SENSITIVITY

C.1 Frozen confirmatory protocol

The confirmatory protocol was serialized and hashed before any new locked-test model outcome was computed. It fixes data roles, methods, grids, repetitions, estimands, uncertainty calculations, and stopping rules. Locked-test loading requires the manifest hash and writes an append-only access record before loading examples. Validation scripts reject a locked-test role, and validation repetitions are excluded from confirmatory intervals.

The primary contrast in each domain is paired final average performance of uniform IEF minus uniform OPG. The secondary factorial comparison is

$$\{\mathrm{OPG}, \mathrm{IEF}\} \times \{\mathrm{uniform}, \mathrm{RDW}\}.$$

All four cells use the same retained examples, and the RDW cells receive the same unit-normalized parameter-gradient directions. Sampled model-Fisher EWC and MAS (Aljundi et al., 2018) are external regularization baselines. Sequential and offline/joint training are references. Every nonzero diagonal importance is rescaled to mean one across trainable parameters, and every method uses the displayed $\lambda/2$ convention.

This estimator-focused factorial keeps the same post-task diagonal-penalty interface. SI's trajectory accumulation and LwF's distillation objective are discussed in Section 2 but are not mixed into that factorial. Paper-facing EWC, IEWC, and IEWC+RDW correspond respectively to archived EF-EWC, IEWC-E/U-IEWC, and IEWC-GSS/GSS-IEWC labels.

Table 5: Frozen confirmatory roles and repetitions. FOMC is excluded because its distributed evaluation and test files are exact duplicates.

Domain	Data roles	Validation runs	Locked runs
ImageNet-R	Stratified 80%/20% split of the original 24,000-example train set for train/locked test; the previously accessed 6,000-example test set is validation only	3 class orders	10 paired class orders
TRACE	Distributed train/eval/test files for C-STANCE, ScienceQA, and NumGLUE-cm; prompt/answer overlaps are zero	3 task orders	6 task orders × 2 optimizer seeds

ImageNet-R uses the fixed λ grid $\{0.01, 0.03, 0.1, 0.3, 1\}$. TRACE uses $10^{-5} \times \{1, 3, 10, 30, 100\}$. Neither grid permits edge expansion.

ImageNet-R uses a pretrained ViT-B/16 with LoRA rank 4 and ten 20-class tasks, trained for two epochs per task with AdamW at learning rate $3 \cdot 10^{-4}$ and batch size 64. Importance uses 128 examples per completed task and RDW cap 64. TRACE uses Qwen2.5-3B-Instruct with LoRA rank 8, two epochs, AdamW at $2 \cdot 10^{-4}$, batch size 2 with gradient accumulation 8, 512 training examples per task, and 64 importance examples per completed task. Data, loader, optimizer, task-order, importance-order, and sampled-Fisher-label seeds are separately recorded and paired where appropriate.

Confirmatory execution uses explicit DataLoader generators, reseeds each task's training stage independently of intervening importance computation, and requires deterministic PyTorch and CUDA kernels. Before restarting validation, two complete same-configuration runs in each domain were compared after removing wall-clock metadata. Their training losses, evaluation matrices, and, for TRACE, generated example-level predictions matched exactly. These duplicate checks are retained for reproducibility auditing but excluded from selection.

For every paired contrast we report the mean and standard deviation of paired differences, a two-sided 95% Student-t interval, the exact two-sided sign-flip randomization p-value when enumerable, the median, and every paired point. These summaries treat task-order/seed pairs as independent; the interval assumes approximately normal paired differences, and sign-flip validity assumes sign exchangeability under the null. The primary interpretation uses a predeclared practical-difference threshold of 0.01—one percentage point on either zero-to-one final-average metric—rather than a binary significance rule. A pre-lock audit found that the frozen protocol listed five secondary contrasts but called them four; the dated clarification applies Holm correction to all five within a domain. Full five-point validation curves, importance time, total run time, and peak CUDA memory are retained.

TRACE's twelve cells form a balanced cross of six task permutations and two model/optimizer seeds, so they are not twelve independent order draws. A further dated pre-lock amendment therefore leaves the twelve-cell primary analysis, practical threshold, five secondary contrasts, and Holm family unchanged, while adding two dependence checks. First, we average each method over the two seeds within an order and recompute all paired summaries over the six order-level units. Second, we average over all six orders within each seed and report the two seed-level paired differences descriptively, without treating a two-unit interval as reliable inference. The balanced construction preserves the overall paired mean. If the run-level and order-level uncertainty readings differ, the more cautious order-level reading governs claims about TRACE robustness. The six orders are the complete frozen permutation set, not a sample supporting extrapolation to an unlimited population of orders.

IEF uses task-relative damping

$$\tau_A = 10^{-2}\,\mathrm{median}\{s_i : s_i > 0\}.$$

Validation-only sensitivity varies the coefficient over $\{0, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$, importance counts, RDW caps, and RDW sample order. A separate loss-unit diagnostic multiplies a fitted classification loss by 0.1, 1, and 10. OPG raw scale changes quadratically, fixed numerical damping changes IEF shape, and task-relative damping preserves the IEF matrix to numerical tolerance, matching Section 11.

C.2 Original multi-domain protocol

Table 6 records the configurations behind Table 2. These values are retained for exact provenance. In the original study, final reported endpoints participated in grid centering, edge expansion, and method selection; the runs therefore do not have an untouched selection-independent endpoint. The original implementation also omitted the displayed factor 1/2, so its numerical λ values should not be compared directly with the revised convention.

Table 6: Configurations for the original multi-domain study. "DR" denotes EWC-DR. These hyperparameters belong to the original implementation and selection workflow.

Task type	Dataset, sequence, and model	Originally selected values
Classification	ImageNet-R; ten 20-class distributions; pretrained ViT-B/16 with LoRA rank 4.	OPG: $\lambda = 333.333$; DR: $\lambda = 3$; U-IEWC: $\lambda = 33.333, \tau = 10^{-4}$; weighted IEWC: $\lambda = 11.111, \tau = 10^{-2}$.
NLP	TRACE, C-STANCE $\to$ FOMC $\to$ ScienceQA; pretrained Qwen2.5-3B with LoRA rank 8.	OPG: $\lambda = 1$; DR: $\lambda = 1$; U-IEWC: $\lambda = 0.111, \tau = 10^{-2}$; weighted IEWC: $\lambda = 59049, \tau = 10^{-2}$.
Forecasting	Weather $\to$ ECL $\to$ Traffic; PatchTST trained from scratch.	OPG: $\lambda = 729$; U-IEWC: $\lambda = 9, \tau = 10^{-2}$; weighted IEWC: $\lambda = 9, \tau = 10^{-2}$.
Segmentation	VOC 15–5, base then novel classes; pretrained ConvNeXtV2-tiny FPN with full fine-tuning.	OPG: $\lambda = 0.111$; DR: $\lambda = 243$; U-IEWC: $\lambda = 270, \tau = 10^{-3}$; weighted IEWC: $\lambda = 2430, \tau = 10^{-2}$.
Diffusion	MNIST DDPM, labels 0/1 $\to$ 2/3; conditional DDPM trained from scratch.	OPG: $\lambda = 9 \cdot 10^5$; U-IEWC: $\lambda = 675, \tau = 10^{-3}$; weighted IEWC: $\lambda = 225, \tau = 10^{-3}$; output ablation: $\lambda = 10^6, \tau = 10^{-3}$.

The original search used a five-point multiplicative grid around a task-specific midpoint and extended an edge up to three times when an endpoint was best. This workflow supports descriptive selected-performance claims but not a selection-independent effect estimate. The revised protocol fixes all five points before validation, forbids edge expansion, and accesses the locked test only after a selection artifact is frozen.

The retained development ledger makes the broader selection history explicit rather than indexing only the five paper-facing settings. It contains 635 per-λ outcomes: 148 vision-classification, 161 forecasting/regression, 47 DDPM, 55 segmentation, 110 encoder-language, and 114 TRACE decoder-only runs. The record includes pilots, smoke checks, tuning sweeps, selected-seed repetitions, and negative or mixed configurations across CIFAR-100 and ImageNet-R; ETT, M4, and long-horizon forecasting; synthetic and GLUE-family language sequences; three Qwen scales on TRACE; VOC 15–5; and MNIST DDPM. It also retains sequential, OPG-EWC, EWC-DR, uniform IEWC, residual-diversity, and FROMP-derived variants where implemented. Thus Table 2 presents attained performance in the strongest representative multi-domain settings, while the archival ledger exposes the full development record and the new locked study supplies selection-independent comparisons.

C.3 Damping and summand diagnostics

Figure 5 combines two complementary damping diagnostics on a common higher-is-better change scale. Panel A re-expresses the historical fixed-τ CIFAR-100/FACIL prefix in changes from $\tau = 10^{-2}$: the two curves are final accuracy and negative forgetting, so values below zero indicate worse performance on either measure. Panel B reports the predeclared relative-κ sweep at the selected λ on one recorded validation repetition per domain, relative to the confirmatory default $\kappa = 10^{-2}$. The default is not uniformly best in these descriptive cells: $\kappa = 1$ changes validation performance by $+0.0263$ on ImageNet-R and $+0.0007$ on TRACE. The former is material under the prespecified one-point threshold while the latter is negligible. These sensitivity outcomes motivate reporting and tuning damping rather than replacing the frozen confirmatory setting post hoc. Fixed τ has loss-dependent units; relative κ is dimensionless and preserves global loss-unit invariance.

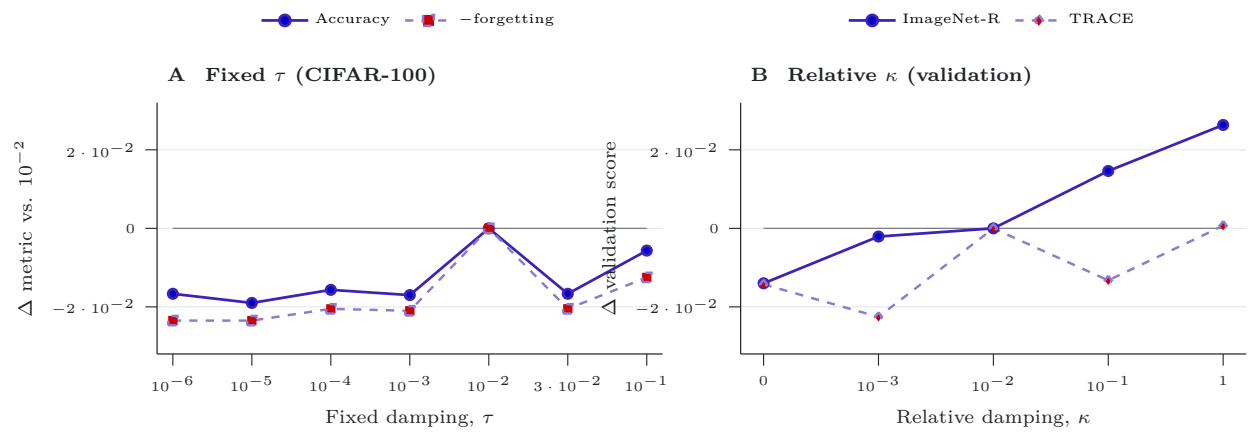


Figure 5: Damping sensitivity. **A**: Historical fixed-τ CIFAR-100 prefix, expressed relative to $\tau = 10^{-2}$; for comparability, forgetting is sign-reversed so higher is better. **B**: Validation-only task-relative damping at fixed selected λ, expressed relative to the predeclared $\kappa = 10^{-2}$ baseline. Panel B uses one prespecified validation repetition per domain and is descriptive rather than a new selection step.

Figure 6 reports the diagonal scale diagnostic for CIFAR-100 classification, controlled regression, MNIST DDPM denoising, and binary segmentation. It shows λ times the per-sample diagonal trace before averaging and is not a sample-weight plot.

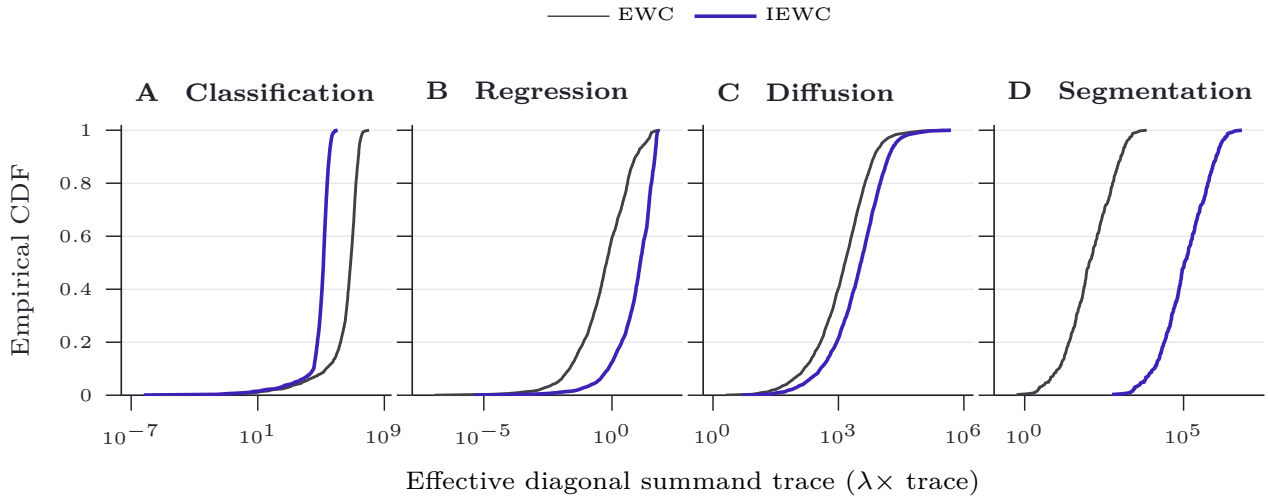


Figure 6: Effective diagonal summand norms. Each panel reports the empirical CDF of λ times the per-sample diagonal trace in the original configuration.

The earlier EWC-DR check additionally tested $\lambda \in \{100, 300, 1000, 3000, 10000\}$ on that CIFAR-100 prefix and selected 100. It remains a local historical sensitivity check, not part of the ImageNet-R confirmatory selection.

D LABEL-NOISE STRESS TEST

The original CIFAR-100 label-noise experiment provides a downstream stress test complementary to the direct matrix analysis in Appendix B. Its endpoints were available during method development, so the table reports descriptive selected performance rather than a selection-independent effect estimate. Uniform training-label flips create atypical high-loss samples, and the clean task-aware test set measures downstream retention. Across this stress test, IEWC loses less accuracy than observed-label OPG-EWC as the flip rate increases.

Table 7: Final average task-aware accuracy under uniform training-label flips, reported as mean ± across-seed standard deviation.

Method	$p = 0$	$p = 0.1$	$p = 0.25$
OPG-EWC	0.5617 ± 0.0062	0.5132 ± 0.0112	0.2773 ± 0.0320
EWC-DR	0.5932 ± 0.0296	0.5310 ± 0.0226	0.4598 ± 0.0095
IEWC	0.5978 ± 0.0260	0.5642 ± 0.0215	0.5322 ± 0.0239

The original study also compared sorted diagonal entries on a CIFAR-100 prefix. That diagnostic does not identify clean and corrupt sample contributions and cannot establish the spectral mechanism by itself; Appendix B now supplies the direct per-sample full-matrix test. The diagonal summary remains useful as a complementary downstream view. It uses all 512 exported entries per matrix and reports the stored-tail mass ratio, normalized stored-tail profile change, and log-scale RMSE of the sorted entries.

Table 8: Sorted-diagonal diagnostics under label noise.

Method	Noise	Entries	Tail mass ratio	Profile L1	Log-scale RMSE
OPG-EWC	0.1	512	0.0288	0.2610	1.4660
OPG-EWC	0.25	512	0.0244	0.2737	1.5337
IEWC	0.1	512	1.0833	0.0350	0.0350
IEWC	0.25	512	1.1003	0.0154	0.0415

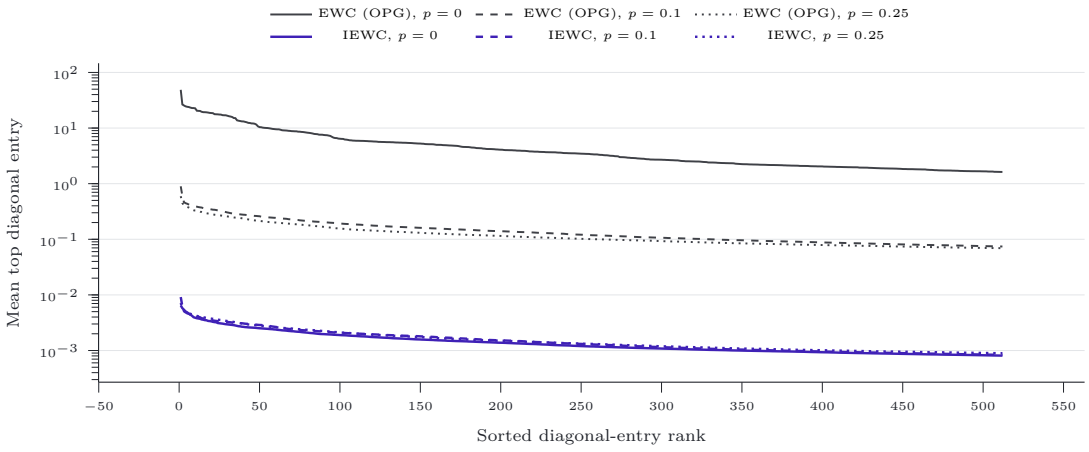


Figure 7: Label-noise diagonal diagnostic. The plot visualizes the sorted high-importance coordinate tail and complements the direct clean/corrupt contribution analysis in Appendix B.

E QUALITATIVE DDPM SAMPLES

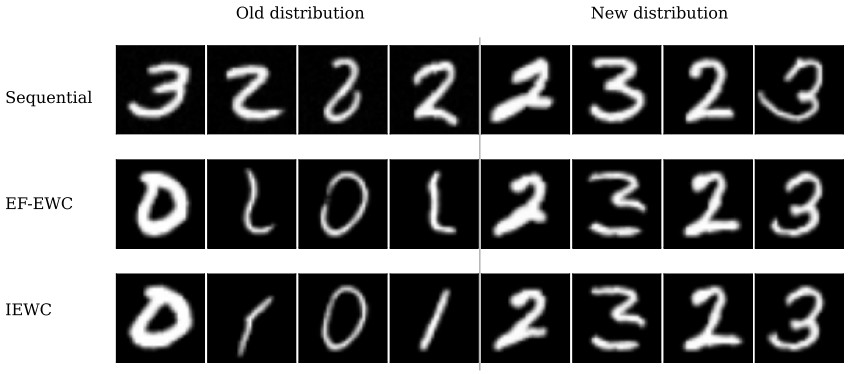


Figure 8: Diffusion generated samples. The grid shows class-conditional MNIST generations from the DDPM architecture after a longer EMA training budget and continual training on both distributions. Rows compare sequential training, OPG-EWC, and IEWC; columns show four old-distribution samples and four new-distribution samples. The old column alternates labels 0 and 1, and the new column alternates labels 2 and 3. Sequential training produces recognizable new-distribution digits but old-label samples drift toward new-distribution digits. OPG-EWC and IEWC preserve old-label generations while learning recognizable new-label samples. This MNIST setting does not support the pattern reported in Rank1Diffusion, where EWC alone struggles in diffusion continual learning and diagonal EWC is ineffective relative to rank-1 Fisher regularization with generative distillation (Wang et al., 2025).

F REPRODUCIBILITY AND ARTIFACT CONTENTS

The supplementary archive supports two distinct reproduction tasks. First, all paper tables, intervals, and figures can be regenerated on CPU from the preserved analysis-complete records without downloading benchmark data or rerunning model training. Second, the model results can be rerun from the frozen source after obtaining the datasets and model checkpoints under their original terms. The archive contains the exact run-source snapshot and Git history, a later reproduction-source snapshot, the manuscript source, the protocol and its amendments, the canonical manifest and selected-lambda artifacts with SHA-256 sidecars, validation and locked-test records, sensitivity records, mechanism diagnostics, reporting scripts, a pinned dependency lockfile, and file-level checksums. A bundle verifier checks counts, paths, hashes, source ancestry, protocol identity, and the absence of restricted example payloads before accepting an extracted archive.

The paper-facing empirical scripts use one `IEWCConfig` interface across the diagonal and low-rank implementations. The same interface records output geometry, damping, aggregation weights, and matrix approximation choices and is used by the frozen experiment and reproduction entrypoints.

Environment and execution. All confirmatory records identify the clean source commit `5e170ea7d84d7094dd1734ee1516b9fba2133a63` and the same canonical protocol-manifest hash. The environment is reconstructed with `uv sync -frozen -dev`; the frozen source passes 75 CPU tests. Runs enable deterministic PyTorch algorithms, disable cuDNN benchmarking and TF32, and separately record model, training-loader, importance-order, and model-Fisher sampling seeds. Each task reseeds training before constructing its data loader, preventing method-specific importance computation from changing a later task's minibatch order. Result files are written atomically. Validation selection rejects missing or duplicate cells, mixed commits, manifest mismatches, and locked records; locked evaluation requires the frozen selected lambda artifact and records access before loading test examples.

The confirmatory suite comprises 192 validation cells, 176 locked-test cells, and 48 validation-only sensitivity cells. ImageNet-R uses ten paired locked class orders. TRACE uses all six permutations of its three retained tasks, crossed with two optimizer seeds; the analysis additionally aggregates within task order to avoid treating the twelve cells as twelve independent order draws. The direct contamination experiment uses twenty independent

seeds at each of four corruption levels. Every prespecified cell is retained regardless of effect direction. The analysis reports paired differences, Student-t intervals, exact sign-flip tests where enumerable, the predeclared practical threshold, and the complete prespecified secondary family.

Data and model provenance. The archive redistributes no benchmark images, language examples, or pretrained weights. ImageNet-R is obtained from its authors; split definitions are pinned to Mammoth commit `e75a491c69fd729edeb01431afb753d9157d9a81`. The frozen roles contain 19,206 training, 6,000 validation, and 4,794 locked-test images across 200 classes. The ViT-B/16 checkpoint is pinned to Hugging Face revision `063c6c38a5d8510b2e57df480445e94b231dad2c`. TRACE is obtained from its authors under the Apache-2.0 repository license. Its distributed train, evaluation, and test files define the three data roles for C-STANCE, ScienceQA, and NumGLUE-cm; the manifest hashes every role and verifies zero prompt and prompt–answer overlap. FOMC is excluded because its distributed evaluation and test files are identical. Qwen2.5-3B-Instruct is pinned to revision `aa8e72537993ba99e69dfaafa59ed015b17504d1` and remains subject to its model terms. Mammoth is MIT licensed, and the pinned `timm` checkpoint declares Apache-2.0. Original IEWC source is supplied for review; the submission does not override third-party terms or redistribute their assets.

Compute and complexity. The confirmatory experiments ran on one workstation with an Intel Core i9-13900K CPU and one NVIDIA GeForce RTX 5090 with 32 GiB of memory. Every run records total wall time, and every importance phase records wall time and peak CUDA allocation. With d trainable parameters and N retained importance samples, diagonal uniform OPG and IEWC store $O(d)$ values and require N per-sample gradients. RDW maintains at most K unit residual directions on CPU, using $O(Kd)$ additional storage and $O(NKd)$ inner-product work before the same diagonal aggregation. The archived efficiency table reports the measured cost of every locked method in both domains.

Regeneration checks. The reproduction guide gives exact commands to rebuild the locked analysis and all generated LaTeX, PDF, PNG, and machine-readable reporting artifacts in a fresh directory. Their checksum manifest is byte-reproducible after removal of volatile PDF timestamps. A clean-checkout audit reconstructs the environment, runs the test suite, regenerates the analysis and paper artifacts, compiles the manuscript, and compares the resulting checksums. Historical multi-domain evidence is retained through the complete 635-outcome ledger and path-level checksums; records that may contain third-party prompts or targets are not redistributed.

G MATCHED ORDERED-CDF OUTPUT GEOMETRY

Consider probability vectors on an ordered grid with m bins. Let

$$P = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$$

project onto the probability-simplex tangent, let $C \in \mathbb{R}^{(m-1)\times m}$ compute cumulative sums through the penultimate bin, and let Δx be the grid spacing. Define

$$G_{\mathrm{CDF}} = \Delta x\, PC^\top CP.$$

For a zero-sum probability perturbation Δz,

$$\Delta z^\top G_{\mathrm{CDF}} \Delta z = \Delta x\, \|C\Delta z\|_2^2,$$

which is the squared L^2 distance between the corresponding one-dimensional cumulative distribution functions. Because G_{CDF} is singular in the ambient space, its use is restricted to the probability-simplex tangent. The matched dual form for a loss covector r is

$$(Pr)^\top G_{\mathrm{CDF}}^{+}(Pr),$$

where G_{CDF}^{+} is the Moore–Penrose pseudoinverse. The implementation tests both the primal identity and the matched dual construction numerically.

This is an ordered-CDF or Cramér metric, not sliced Wasserstein. It supplies one explicit space-dependent geometry compatible with IEWC's matched primal/dual construction. The retained diffusion output-representation experiment predates this correction and is therefore treated only as exploratory evidence that alternative output representations can affect the penalty, not as empirical validation of Wasserstein geometry.